Scale Repline Proposal

Each Bond has a number of pools/loans as collateral. The particular distribution of characteristics: CLTV, Incentive, FICO, LoanSize, etc. will influence the model projections and hence the price/OAS valuations.

Scale currently has prepayment models calibrated to pool level data, so we will keep our discussion of repline algorithms to the pool level collateral. Most of the arguments outlined in this document extend to loan level models and collateral.

The goal for any repline algorithm is dimension reduction which facilitates speed of calculation, while at the same time capturing the distribution of the underlying collateral as far as it has an effect on the model projections.

For example, a Bond might have 20 pools all with very different CLTV values. This could be a good variable to bucket by because CLTV itself has high dispersion. However, if all the CLTV values are in the 0-65 range, then the model will not differentiate much in terms of CPR prediction as impacted by the CLTV variable. If however, the CLTV range is from 78-110, then the model predicted CPR will vary considerably based on these CLTV value. Hence, the choice of collateral variable to bucket by in our repline algorithm is the combination of: 1) collateral variable distribution, and 2) model impact on the collateral variable.

In the algorithm that we propose, we will restrict ourselves to 1 dimension (1 variable) for bucketing. For example, we can bucket by CLTV, or by FICO, but not by both.
We also restrict ourselves to 1 period of projections. The algorithm could be extended for multiple dimensions of bucketing and multiple periods in future versions.

The goal for this 1 dimension, 1 period algorithm is that for any bond we can determine the rank order of collateral variables as measured by how much of the Model CPR dispersion it captures. The highest ranked collateral variable for a bond would be the collateral variable that we bucket by to create our replines.

The algorithm is as follows:

1) Pull all the pool data for a particular bond for the latest factor date. The data should contain all the required fields for the model to run.
2) Apply the full model to each pool. Capture the intermediate S curve SMM, the multipliers for each model variable, and the total mode prediction for SMM.
3) Calculate a weighted average value for each intermediate model variable. ex) Weighted Average CLTV multiplier.
4) Loop over each collateral variable which has an impact on the total SMM from the model. For each variable, we will calculate a total SMM for each pool using the Weighted Average values for everything except for the model variables which are directly impacted

by the particular collateral variable we are investigating. This effectively gives us a partial derivative of Total SMM as the collateral variable changes from pool to pool.

5) When calculations are done for each collateral variable and for each pool, we calculate a weighted standard deviation for each collateral variable.
6) We rank each collateral variable based on standard deviation from largest to smallest. The collateral variable with the largest standard deviation is the most important variable for us to bucket by in our repline algorithm.
7) We repeat this algorithm for all bonds that we are interested in running.

Example Output:

Bond: G14.21(GS)

Rank	Variable	Standard Deviation
1	**incentive**	**0.002835127**
2	burnout	0.001484082
3	wacls	0.001420036
4	fico	0.000791386
5	wala	0.000188906
6	pct_DELQ	0.000184536
7	cltv	0.000103764

Bond: G14.21(GS)

Rank	Variable	Standard Deviation
1	**fico**	**0.006702**
2	wala	0.003598
3	cltv	0.00152
4	pct_owner	0.001089
5	pct_DELQ	0.000417
6	incentive	0.000227
7	wacls	2.32E-05
8	burnout	1.26E-05

For full examples please see:

Q:\Agency Prepayment Model\Repline Strategy\ginnie_g1421_gs_pool_analysis_v2.xlsx
Q:\Agency Prepayment Model\Repline Strategy\highltv_fhlpc339_ss_pool_analysis_v2.xlsx